Route Profitability & Fleet Efficiency - Hiring Manager Report

A business-readable explanation of the project, pipeline, analytics findings, charts, and next-quarter recommendation.

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Project stack	PySpark, Delta Lake, SQL, dimensional modeling, analytics storytelling
Dataset grain	12,000 route-days; one row per route per day from 2022-01-01 to 2024-12-31
Business goal	Build a repeatable profitability view sliceable by date, region, BU and area

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1. Executive Summary

This project solves a commercial route profitability problem for GFL Environmental. Leadership can see revenue, but revenue alone does not show whether a route is actually profitable after disposal, fuel, labour, maintenance and admin costs are included. The project builds a repeatable PySpark + Delta Lake pipeline that turns route-day data into trusted profitability metrics and BI-ready tables.

- Dataset: 12,000 route-days from 2022 to 2024, covering 120 routes, 29 areas, 8 business units and 6 regions.
- Overall result: $99.6M revenue, $50.7M cost, $48.9M gross profit and 49.1% weighted gross margin.
- Underperformance definition: a route-day below 15% gross margin; persistent routes are flagged using weighted margin and underperforming-day share.
- Main business finding: low-margin performance is mainly a pricing and disposal economics issue, not a broad fuel or labour productivity issue.
- Recommendation: run a targeted Margin Recovery Sprint for General Waste / Office & Commercial routes in Airdrie, Laval, Kingston, Toronto East and Toronto North.

2. Business Problem and Project Objective

The business problem is to identify which routes, areas and business units are making money after all major route-level costs are included. Some routes can look attractive because they generate high revenue, but the same routes may still lose margin when disposal tonnage, cost per stop or service complexity is considered.

> **Hiring-manager takeaway:** This is not only a coding exercise. It demonstrates business understanding, metric selection, pipeline design, data quality thinking, and the ability to convert analytics into an operational recommendation.

Project requirement	How the project addresses it
Reliable route-level profitability view	The Silver layer creates one clean row per route-day with profitability and efficiency metrics.
Sliceable by region, BU and area	The Gold model stores monthly route-area data and uses dimensions for date, geography, route, waste stream and customer segment.
Underperforming route definition	A route-day under 15% gross margin is flagged; persistent underperformance is evaluated using weighted margin and frequency of underperforming days.
Data-backed recommendation	The analysis isolates weak areas, waste streams and customer segments, then recommends focused pricing and disposal actions.

3. What I Built Technically

The solution uses a medallion architecture because it separates raw ingestion, metric logic and BI consumption. This is important in a real data engineering environment because raw data can be corrected later, business metric definitions must be repeatable, and dashboard users need fast, clean tables.

Pipeline architecture

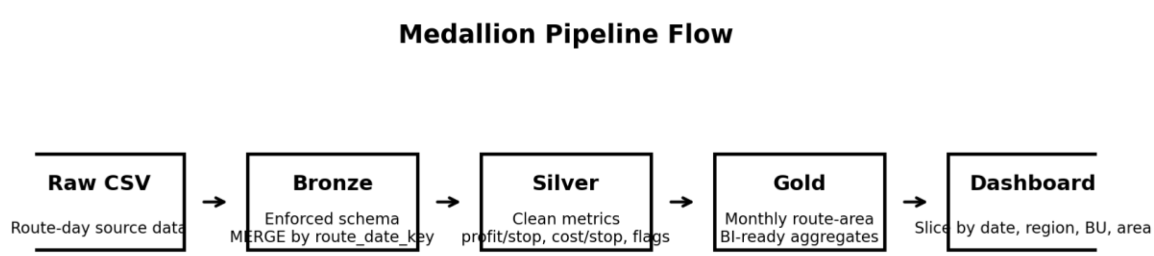

Figure 1. Bronze, Silver and Gold pipeline flow.

Graph insight: Bronze protects the raw source and audit trail, Silver owns metric definitions, and Gold provides stable aggregates for dashboards. This makes the project production-oriented instead of only exploratory.

Layer	Purpose	Important design choices
Bronze	Load raw route-day CSV with enforced schema and ingestion metadata.	Avoid inferSchema, keep source_file and ingestion_ts, use MERGE by route_date_key for late corrections.
Silver	Clean route-day data and create reusable financial and operational metrics.	Create revenue_per_stop, cost_per_stop, profit_per_stop, disposal_cost_share, fuel_l_per_100km, stops_per_labour_hour, completion_rate and flags.
Gold	Build dashboard-ready route-month and area/BU tables.	Partition by year/month and use ZORDER by region, BU, area and route_id for slicer performance.

4. Data Quality and Metric Trust

A hiring manager will care about whether the numbers can be trusted. The project includes checks for row uniqueness, date range, numeric nulls, cost reconciliation and incident type logic. Cost fields reconcile because total_cost equals disposal + fuel + labour + maintenance + admin cost, and gross_profit equals total_revenue minus total_cost.

Quality check	What it proves
12,000 rows loaded	The full supplied dataset is captured.
Unique route_date_key	The project preserves one authoritative row per route-day.
Date range 2022-01-01 to 2024-12-31	The trend analysis covers the expected three-year window.
Numeric KPI nulls = 0	Core financial and operational calculations can run without missing-value distortion.
Cost reconciliation	Total cost equals the component cost sum except small floating-point rounding noise.
Revenue reconciliation	Average revenue per stop x completed stops approximately equals total revenue, with small differences due to rounded averages.

5. Metrics Selected and Why They Matter

Metric	Formula / logic	Why a manager cares
Weighted gross margin %	gross_profit / total_revenue	Main profitability KPI; weighted rollups avoid misleading simple averages.
Profit per stop	gross_profit / completed_stops	Simple route economics measure that managers can understand quickly.
Revenue per stop	total_revenue / completed_stops	Shows pricing strength and customer mix.
Cost per stop	total_cost / completed_stops	Shows normalized cost pressure independent of route size.
Disposal cost per tonne	disposal_cost / total_tonnes	Identifies expensive disposal burden and material-mix issues.
Stops per labour hour	completed_stops / labour_hours	Measures crew productivity.
Fuel L / 100 km	fuel_litres / distance_km x 100	Measures fleet efficiency.
Completion rate	completed_stops / total_stops	Measures service execution quality.

6. Key Findings with Graph Insights

Finding 1 - Disposal cost dominates cost structure

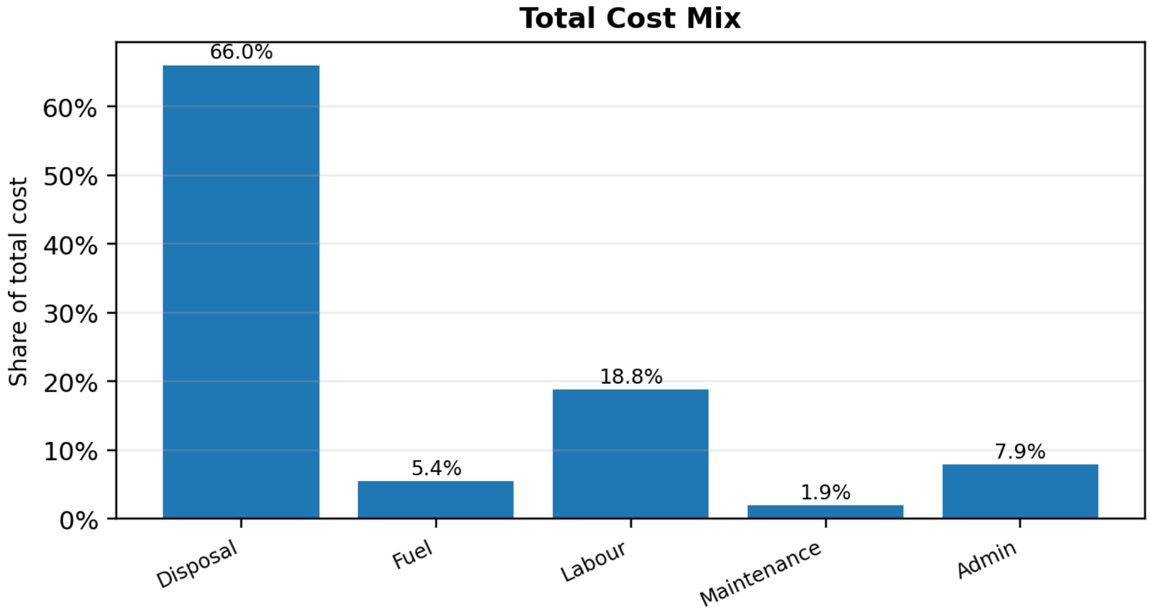

Figure 2. Total cost mix across all route-days.

Graph insight: Disposal is 66.0% of total cost, far larger than labour at 18.8%, admin at 7.9%, fuel at 5.4% and maintenance at 1.9%. This tells the hiring manager that the main operational lever is not only fuel or labour; disposal economics must be reviewed.

Finding 2 - Profitability improved over time

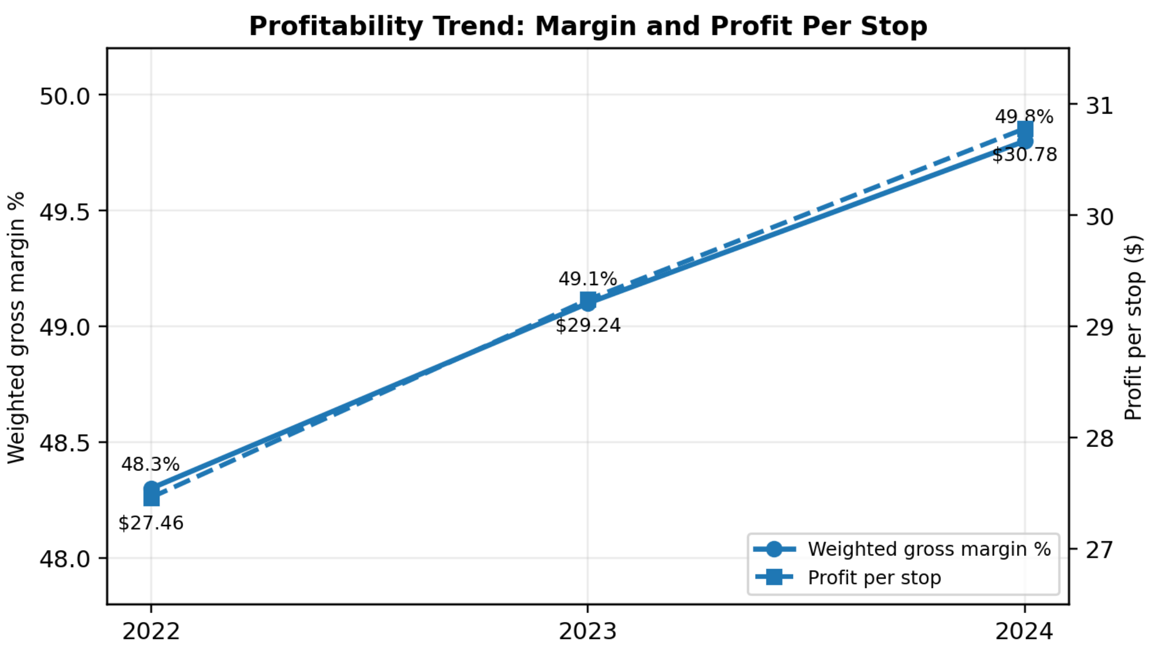

Figure 3. Weighted margin and profit per completed stop, 2022-2024.

Graph insight: Weighted gross margin increased from 48.3% to 49.8%, while profit per stop rose from $27.46 to $30.78.

Revenue per stop increased faster than cost per stop, so the business is improving overall even though pockets of underperformance remain.

Finding 3 - Calgary BU is the weakest business unit

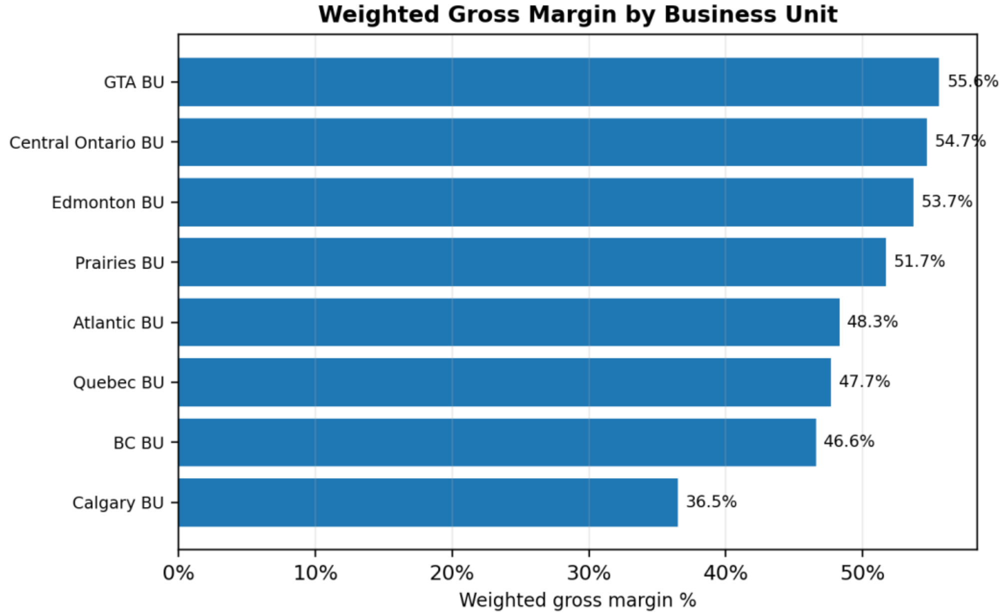

Figure 4. Weighted gross margin by business unit.

Graph insight: Calgary BU is the clearest BU-level concern at 36.5% margin and 19.8% underperforming route-days. Atlantic BU also needs monitoring because it has the highest negative route-day rate even though its aggregate margin is closer to average.

Finding 4 - Airdrie, Laval and Kingston need urgent action

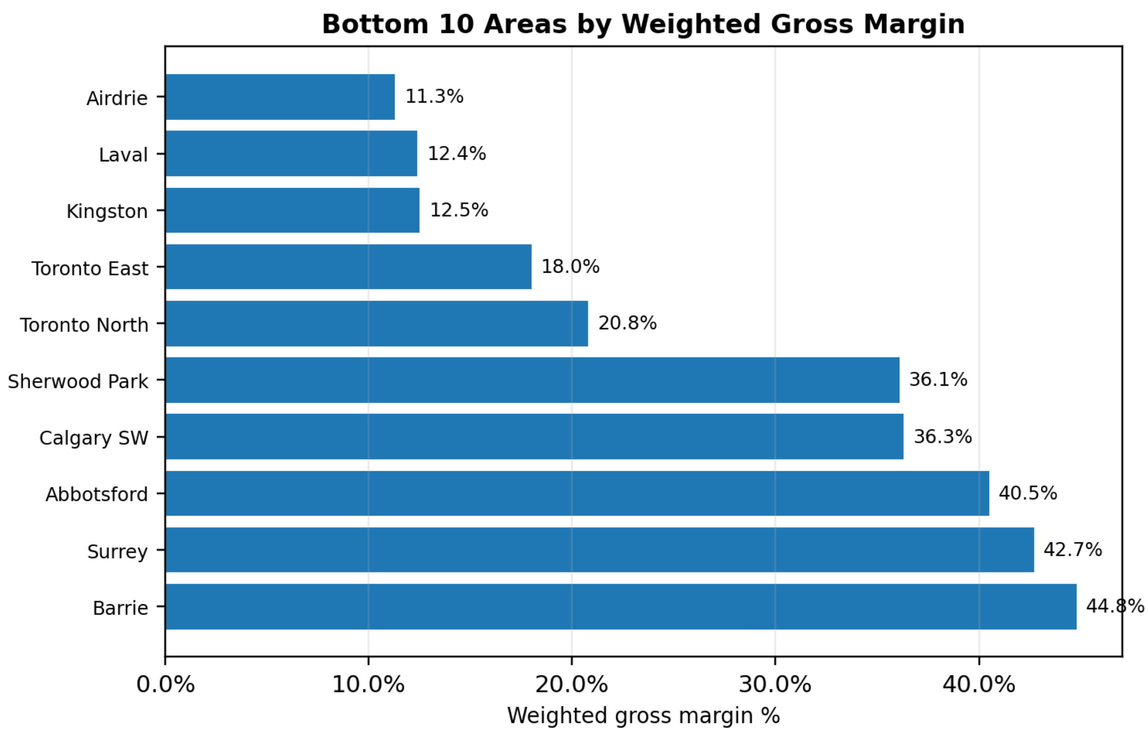

Figure 5. Lowest-margin areas by weighted gross margin.

Graph insight: Airdrie, Laval and Kingston are not small-sample outliers. They have 102-200 route-days, margins near 11-13%, and more than 60% of route-days below the 15% threshold. These are practical targets for next-quarter intervention.

Finding 5 - Underperformance is caused by price/disposal economics

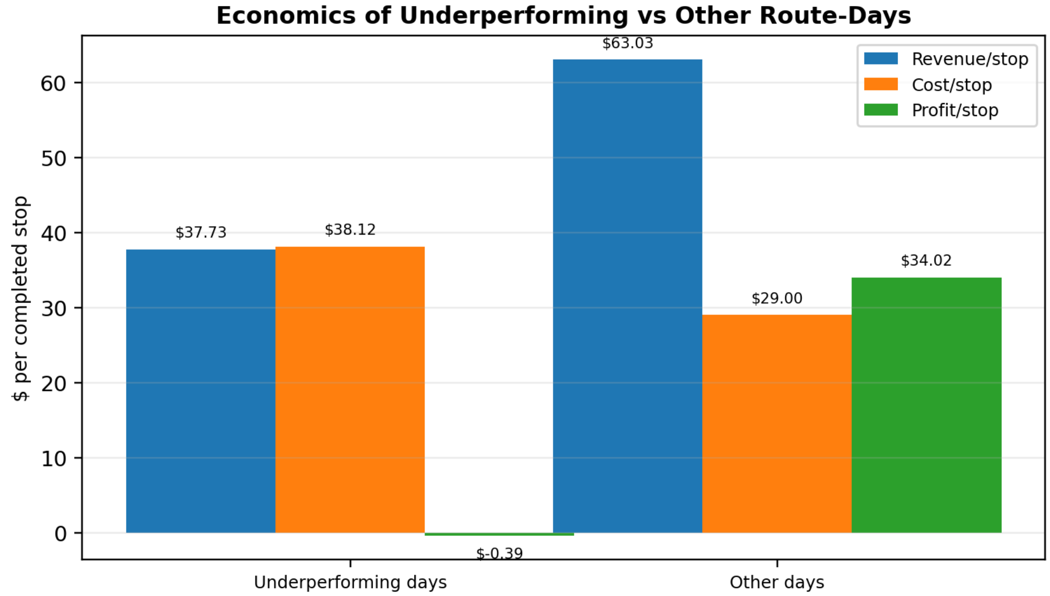

Figure 6. Route-day economics: underperforming vs other days.

Graph insight: Underperforming days have much lower revenue per stop ($37.73 vs $63.03) and higher cost per stop ($38.12 vs $29.00), producing almost zero or negative profit per stop. Fuel efficiency and labour productivity are nearly similar, so the root cause points to pricing, customer mix, tonnage and disposal burden.

Finding 6 - General Waste is the weakest waste stream

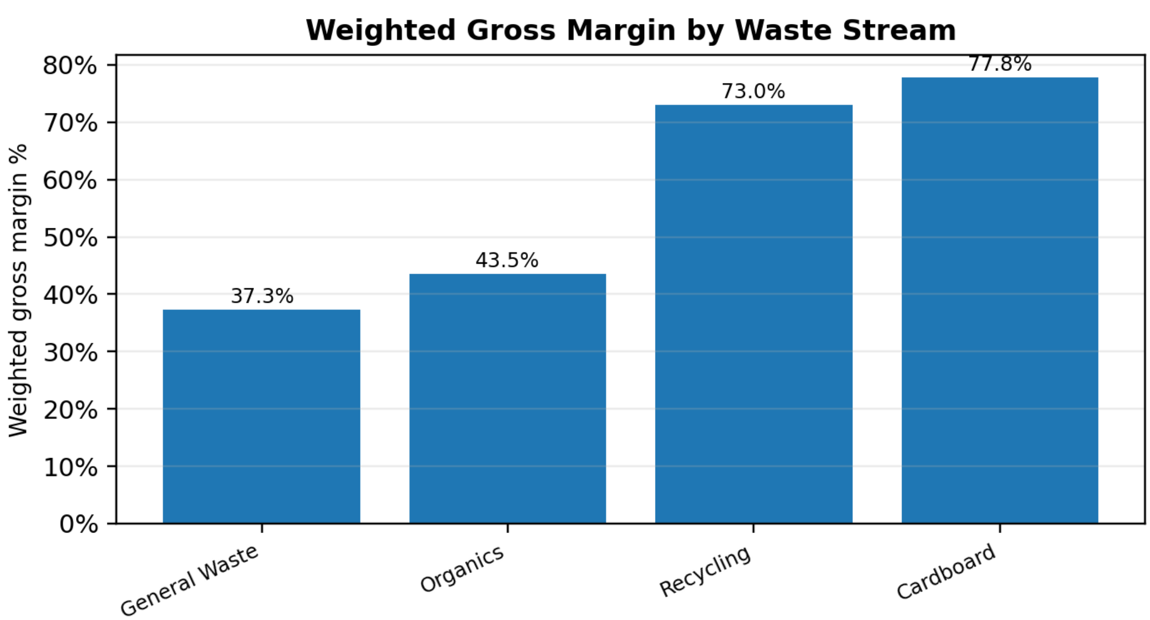

Figure 7. Waste-stream profitability comparison.

Graph insight: General Waste has the weakest weighted margin at 37.3%, while Recycling and Cardboard are above 70%. This helps justify a focused recommendation rather than applying broad cost cuts across all routes.

Finding 7 - Office & Commercial is the weakest customer segment

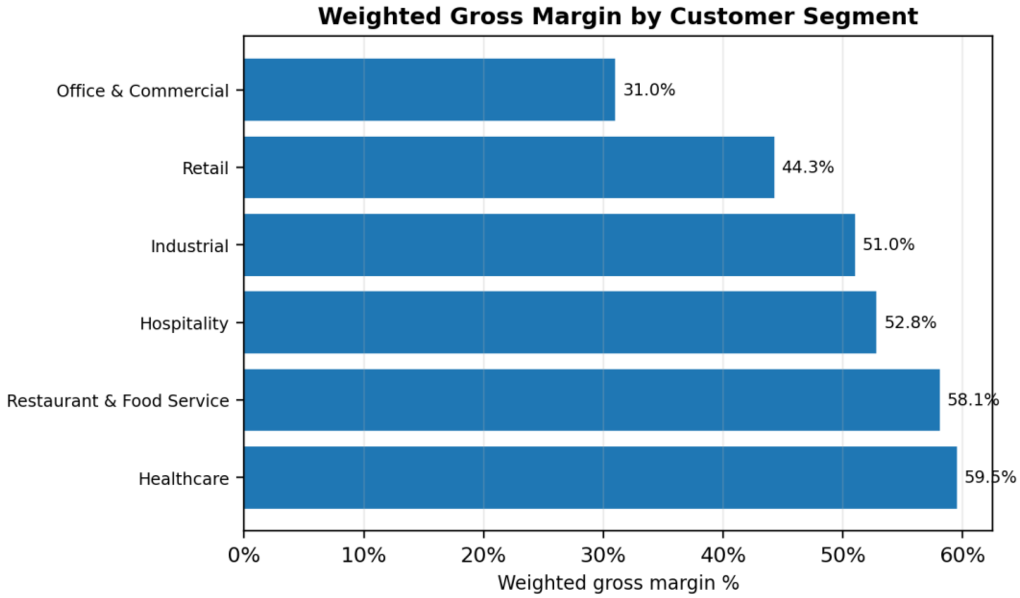

Figure 8. Customer segment profitability comparison.

Graph insight: Office & Commercial has the weakest margin at 31.0% and the lowest profit per stop. In the weak areas, the issue is strongly tied to General Waste or Organics service for Office & Commercial customers.

7. Underperformance Logic

The underperformance threshold is intentionally business-friendly. A route-day below 15% gross margin is flagged because the median route-day margin is 47.0%, the 25th percentile is 28.6%, and the 10th percentile is 9.7%. A 15% floor is severe enough to identify serious issues without flagging every normal daily variation.

Threshold support	Value
P10 route-day margin	9.7%
P25 route-day margin	28.6%
Median route-day margin	47.0%
Chosen underperformance threshold	< 15.0% gross margin
Route-days below threshold	1,592 (13.3%)
Negative margin route-days	717 (6.0%)

8. Data Model for Dashboarding

The dimensional model supports a route profitability dashboard with slicers for date, region, BU, area, route, waste stream and customer segment. The core fact table stays at route-day grain because this preserves detail and allows flexible aggregation.

Dimensional model

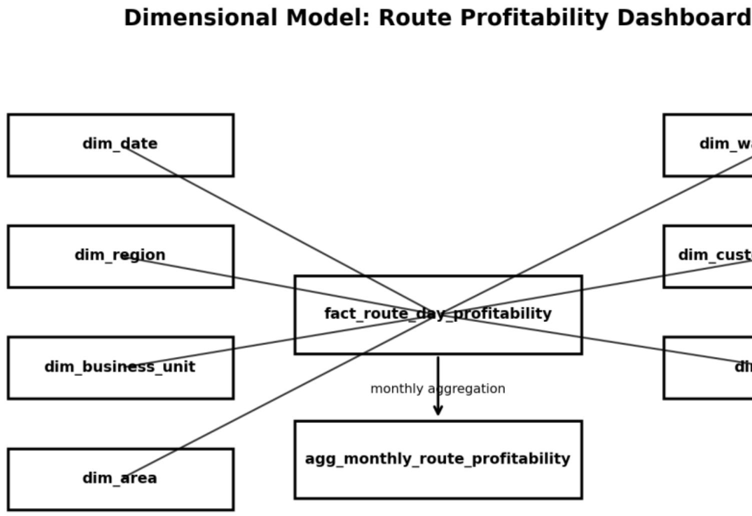

Figure 9. Star-style model for route profitability reporting.

Graph insight: The model separates facts from descriptive dimensions. This makes Power BI or SQL dashboard queries easier, enables reusable definitions, and supports business-friendly slicing without exposing raw pipeline complexity.

Table	Grain	Purpose
dim_date	One row per calendar date	Time slicers for year, month, quarter and day of week.
dim_region	One row per region	Top-level geography reporting.
dim_business_unit	One row per BU	BU hierarchy and rollup to region.
dim_area	One row per area	Area hierarchy and local operational analysis.
dim_route	One row per route_id	Route entity and descriptive attributes.
dim_waste_stream	One row per waste stream	Waste stream margin comparison.
dim_customer_segment	One row per customer segment	Customer segment profitability analysis.
fact_route_day_profitability	One row per route per day	Core financial, operational, safety and schedule measures.
agg_monthly_route_profitability	One row per month-route-area-waste-segment	Gold table optimized for dashboard queries.

9. Data-Backed Recommendation

Launch a next-quarter Margin Recovery Sprint focused on General Waste / Office & Commercial routes in Airdrie, Laval, Kingston, Toronto East and Toronto North. This recommendation is targeted because the analysis shows the issue is concentrated by area, waste stream and customer segment.

Action	What to do	Why it matters
Pricing review	Audit contracts and customers with revenue per stop below area peers.	Low revenue per stop is the biggest difference between underperforming and other route-days.
Disposal review	Review disposal site selection, contamination, container mix and tonnage patterns.	Disposal is the largest cost driver and is much higher on underperforming days.
Route operations	Check stop density, pickup frequency, route sequencing and missed-stop causes.	Labour and fuel are not the main overall driver, but route-level operational fixes can still support recovery.
Monthly measurement	Track margin %, profit per stop, disposal cost per tonne, underperforming-day rate and recovery dollars.	This turns the analysis into an ongoing management process instead of a one-time report.